WW-T-700/5E
29 December 1983
SUPERSEDING
WW-T-700/5D
March 13, 1967

FEDERAL SPECIFICATION SHEET

TUBE, ALUMINUM ALLOY, DRAWN, SEAMLESS, 5086

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

The complete requirements for procuring seamless tube drawn from aluminum alloy 5086 described herein shall consist of this document and the latest issue of WW-T-700/GEN (see 2.1).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for seamless tube drawn from aluminum alloy 5086.

1.2 Classification.

1.2.1 Tempers. The drawn seamless tube shall be of the following tempers: O, H32, H34, H36, and F, as specified (see 6.2 and 6.3). The definitions of these tempers shall be as specified in WW-T-700/GEN.

1.2.2 Types. The tube shall be of the following types:

Type		Appearance
I	-	Round
II	-	Rectangular and square
III	-	Streamline
IV	-	Oval
V	-	Odd shapes

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specification

WW-T-700/GEN - Tube, Aluminum and Aluminum Alloy, Drawn, Seamless, General Specification for

FSC 4710

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3. REQUIREMENTS

3.1 Chemical composition. The chemical composition shall conform to the requirements specified in table I.

TABLE I. Chemical composition 1/

Element	Percent	
	Minimum	Maximum
Silicon	-	0.40
Iron	-	0.50
Copper	-	0.10
Manganese	0.20	0.7
Mangesium	3.5	4.5
Chromium	0.05	0.25
Zinc	-	0.25
Titanium	-	0.15
Others, each	-	0.05
Others, total	-	0.15
Aluminum	Remainder	

1/ Except for "Aluminum" and "Others", analysis normally is made for elements for which specific limits are shown.

2/ The sum of those "Others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum

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3.2 Mechanical properties.

3.2.1 Tensile strength, yield strength and elongation properties. The tensile strength, yield strength and elongation mechanical properties parallel to the direction of drawing shall conform to the requirements specified in table II.

TABLE II. Tensile strength, yield strength and elongation properties

Temper	Wall thickness, inch	Tensile strength, minimum ksi	Yield strength, minimum, ksi	Percent elongation in 2 inches, minimum
O	0.010-0.450 incl.	35.0 1/	14.0	14
H32	0.010-0.050 incl. 0.051-0.450 incl.	40.0 40.0	28.0 28.0	6 8
H34	0.010-0.050 incl. 0.051-0.450 incl.	44.0 44.0	34.0 34.0	5 6
H36	0.010-0.050 incl. 0.051-0.450 incl.	47.0 47.0	38.0 38.0	4 5
F	All	2/	2/	2/

1/ Maximum of 46.0 ksi
2/ No requirements

3.2.2 Flattening. When specified (see 6.2), round tube (type 1) in O and H32 tempers shall withstand, without cracking, the flattening test or the alternative bend test specified in WW-T-700/GEN. The values for flattening factor "F" are specified in table III.

TABLE III. Flattening factor

Temper	Wall thickness, inch	F
O	0.010 to 0.450, incl.	3
H32	0.010 to 0.450, incl.	8

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3.2.2.1 Alternative bending factor "N". The values for the alternative bending factor "N" are specified in table IV.

TABLE IV. Bending factor

Temper	N
O	1
H32	6

3.2.3 Leak. When specified (see 6.2), round tube (type I) shall withstand either the pressure test or the electromagnetic (eddy current) test specified in WW-T-700/GEN.

4. QUALITY ASSURANCE PROVISIONS (see WW-T-700/GEN)

5. PREPARATION FOR DELIVERY (see WW-T-700/GEN)

6. NOTES

6.1 Intended use. This tube is intended for use when good strength and weldability are required. This tubing is not applicable for use in transferring hydraulic/pneumatic mediums.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number and date of this specification
(b) Temper of material required (see 1.2.1)
(c) Type of tube required (see 1.2.2)
(d) Whether round tube (type 1) in O and H32 tempers should withstand flattening (see 3.2.2)
(e) Whether round tube (type I) should withstand leak test (see 3.2.3)
(f) Dimensions required
(g) Requirements for sizes not specifically covered (see WW-T-700/GEN)
(h) Selection of applicable levels of preservation and packing (see WW-T-700/GEN)

6.3 Availability of the types. Types II through V are only available in the O and F tempers. Type I is available in all tempers.

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Review Activities

Army-AR
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User Activities

Army-MI
Navy-MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
COMMERCE-NBS
HEW-FEC
NASA-KSC, MSF
USDA-AFS

PREPARING ACTIVITY:

NAVY-AS

DoD Project 4710-0711

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
WW-T-700/5E	TUBE, ALUMINUM ALLOY, DRAWN, SEAMLESS, 5086

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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